

Escaping the Pentane Barrier

Conformational Sampling of *n*-Pentane Using Baseline Monte Carlo,
NVT Molecular Dynamics, and Umbrella Sampling with WHAM
Using the TraPPE-UA Force Field

Shreyas Deb

Roll No. 2023113024

Molecular Modelling and Simulations

May 2026

Contents

1	Introduction	3
2	Force Field and System Description	3
2.1	United-Atom Representation	3
2.2	Potential Energy Function	3
2.3	Parameter Summary	5
3	Task 1: Initial Configuration	5
3.1	All-Trans Geometry Construction	5
3.2	Coordinate Convention	5
4	Task 2: Baseline Sampling — MC and NVT MD	6
4.1	Monte Carlo Implementation	6
4.2	NVT Molecular Dynamics Implementation	6
4.2.1	Equations of Motion	6
4.2.2	Integration Scheme	7
4.2.3	Force Evaluation	7
4.2.4	Initial Conditions	7
4.3	Simulation Parameters	8
4.4	Results	8
4.4.1	Quantitative Analysis of Trapping	10
5	Task 3: Enhanced Sampling — Umbrella Sampling with WHAM	10
5.1	Motivation	10

5.2	Umbrella Sampling Design	11
5.3	Within-Window Monte Carlo	11
5.4	WHAM Post-processing	11
5.4.1	WHAM Equations	11
5.4.2	Iterative Solution	12
5.4.3	PMF Reconstruction	12
5.5	Results	12
6	Task 4: Sampling Efficiency and PMF Analysis	14
6.1	Exploration Entropy	14
6.2	Early Exploration Score	14
6.3	Potential of Mean Force from Baseline Simulations	14
6.4	Quantitative Results	15
6.5	PMF Features and Physical Interpretation	16
7	Discussion	17
7.1	Why Baseline Methods Fail	17
7.2	Why Umbrella Sampling Succeeds	17
7.3	Consistency between WHAM and Baseline PMF	17
7.4	TraPPE-UA Fidelity	18
8	Conclusions	18

1 Introduction

Molecular simulations are routinely used to explore the conformational landscape of flexible molecules. However, a system that evolves according to physically accurate equations of motion can become indefinitely trapped in a single local free-energy minimum whenever the energy barriers separating minima are large compared to the thermal energy $k_{\text{B}}T$. This “trapping” problem is one of the central challenges in computational statistical mechanics and motivates the development of enhanced sampling strategies [Frenkel & Smit, 2002].

n-Pentane (C_5H_{12}) is a canonical model system for studying torsional trapping. Its central C–C–C–C backbone dihedral φ_1 (defined by atoms C1–C2–C3–C4) partitions conformational space into three well-known minima: the *trans* state ($\varphi_1 \approx \pm 180^\circ$) and the two *gauche* states ($\varphi_1 \approx \pm 60^\circ$). The energy barriers between these states are large enough to induce trapping at low temperature even in simulations of 200 000 steps.

This report presents a complete Python implementation of:

1. An all-atom initial geometry generated from internal coordinates (Task 1);
2. Baseline Metropolis Monte Carlo (MC) and Nosé–Hoover NVT molecular dynamics (MD) simulations at 120 K and 250 K (Task 2);
3. Umbrella sampling with Weighted Histogram Analysis Method (WHAM) as the enhanced sampling strategy that overcomes the barriers (Task 3);
4. Entropy-based sampling efficiency metrics and potential of mean force (PMF) analysis (Task 4).

All simulations use the TraPPE-UA (united-atom) force field of Martin & Siepmann 1998. No external MD engine is used; every algorithm is coded from scratch in Python with NumPy, with Numba JIT compilation used selectively for performance.

2 Force Field and System Description

2.1 United-Atom Representation

In the TraPPE-UA model, hydrogen atoms are merged into the attached carbon centre. *n*-Pentane is therefore represented as five interaction sites: two terminal CH_3 groups (C1, C5) and three internal CH_2 groups (C2, C3, C4). This reduces the number of degrees of freedom while preserving the essential torsional behaviour.

2.2 Potential Energy Function

The total potential energy is the sum of four contributions:

Bond stretch.

$$U_{\text{bond}} = \sum_{\text{bonds}} \frac{1}{2} k_r (r - r_0)^2 \quad (1)$$

TraPPE-UA specifies *rigid* bonds ($r_0 = 1.54 \text{ \AA}$); a stiff harmonic spring with $k_r = 452,900 \text{ K \AA}^{-2}$ (from Mundy et al. 1996) is used here to allow full Cartesian dynamics while maintaining near-rigid bond lengths. At equilibrium geometry, the bond stretch contribution to the total energy is negligible ($\ll 1 \text{ K}$).

Angle bend.

$$U_{\text{angle}} = \sum_{\text{angles}} \frac{1}{2} k_\theta (\theta - \theta_0)^2 \quad (2)$$

with equilibrium angle $\theta_0 = 114.0^\circ$ and force constant $k_\theta = 62,500 \text{ K rad}^{-2}$ from Martin & Siepmann 1998. There are three C–C–C angles in *n*-pentane (C1–C2–C3, C2–C3–C4, C3–C4–C5), all summed over in the implementation.

Torsion. The three-term OPLS-style torsional potential:

$$U_{\text{tors}}(\varphi) = C_0 + C_1(1 + \cos \varphi) + C_2(1 - \cos 2\varphi) + C_3(1 + \cos 3\varphi) \quad (3)$$

with $C_0 = 0 \text{ K}$, $C_1 = 355.03 \text{ K}$, $C_2 = -68.19 \text{ K}$, $C_3 = 791.32 \text{ K}$ from Table 2 of Martin & Siepmann 1998. This potential is summed over both backbone dihedrals φ_1 (C1–C2–C3–C4) and φ_2 (C2–C3–C4–C5). The 1–4 non-bonded interactions are absorbed into these torsion coefficients, so no separate 1–4 LJ terms are added.

Lennard-Jones (1–5 pair only).

$$U_{\text{LJ}}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \quad (4)$$

For *n*-pentane, the exclusion rules are: 1–2 (bonded), 1–3 (angle-separated), and 1–4 (dihedral, absorbed in torsion) pairs are all *excluded*. The only permitted intramolecular LJ interaction is the C1 $\cdots$ C5 (1–5) pair. Parameters: $\epsilon_{\text{CH}_3}/k_B = 98.0 \text{ K}$, $\sigma_{\text{CH}_3} = 3.75 \text{ \AA}$ (both sites are CH₃, so Lorentz–Berthelot combining rules are trivial).

Torsional barrier height. Evaluating Eq. (3) at the cis saddle point ($\varphi = 0$) gives a barrier of approximately 2,292 K. The ratio $\Delta U/k_B T$ equals 19.1 at 120 K and 9.2 at 250 K. The spontaneous barrier-crossing probability under Metropolis sampling is therefore $e^{-19.1} \approx 5 \times 10^{-9}$ at 120 K and $e^{-9.2} \approx 10^{-4}$ at 250 K. For a 200 000-step simulation this predicts essentially zero crossings at 120 K and only ~ 20 at 250 K — motivating the enhanced sampling in Task 3.

2.3 Parameter Summary

Table 1: TraPPE-UA parameters for *n*-pentane. Sources: [Martin & Siepmann \[1998\]](#) (torsion, LJ, angle, bond length); Mundy et al. 1996 (bond spring constant).

Term	Parameter	Value	Unit
Bond length	r_0	1.54	Å
Bond spring	k_r	452 900	K Å ⁻²
Angle	θ_0	114.0	deg
Angle spring	k_θ	62 500	K rad ⁻²
Torsion	$C_1/C_2/C_3$	355.03 / -68.19 / 791.32	K
LJ (CH ₃)	ε/k_B	98.0	K
LJ (CH ₃)	σ	3.75	Å
LJ (CH ₂)	ε/k_B	46.0	K
LJ (CH ₂)	σ	3.95	Å

3 Task 1: Initial Configuration

3.1 All-Trans Geometry Construction

The all-trans starting geometry is constructed analytically from internal coordinates using the standard z-matrix to Cartesian conversion. Beginning from C1 at the origin and C2 placed along the positive x -axis, each subsequent atom is placed using `geometry.place_atom`, which enforces the IUPAC dihedral convention ($\varphi = 0$ is cis, $\varphi = \pm\pi$ is trans). The resulting five-atom coordinate array has all backbone dihedrals set exactly to 180°, confirming all-trans initialisation.

The function `build_all_trans(cfg)` reads all bond lengths and equilibrium angles from the TOML configuration file at runtime. No geometry parameters are hard-coded in the Python source. Self-consistency of the geometry builder is verified by the round-trip test:

$$\text{calc_dihedral}(*\text{build_pentane}(\varphi, \varphi, \text{cfg})[[0,1,2,3]]) \approx \varphi$$

to within floating-point tolerance for any input dihedral φ .

3.2 Coordinate Convention

In this work, all simulations propagate full Cartesian coordinates (x_i, y_i, z_i) for each of the five united-atom sites. The dihedral φ_1 is extracted from the Cartesian positions at every step using `geometry.calc_dihedral`, which computes the signed angle between the planes defined by C1–C2–C3 and C2–C3–C4 via the cross-product method:

$$\varphi_1 = \text{atan2}(\hat{\mathbf{b}}_2 \cdot (\mathbf{n}_1 \times \mathbf{n}_2), \mathbf{n}_1 \cdot \mathbf{n}_2) \quad (5)$$

where $\mathbf{n}_1 = \mathbf{b}_1 \times \mathbf{b}_2$, $\mathbf{n}_2 = \mathbf{b}_2 \times \mathbf{b}_3$, and $\mathbf{b}_k$ are the bond vectors. This gives a result in $(-\pi, \pi]$.

4 Task 2: Baseline Sampling — MC and NVT MD

4.1 Monte Carlo Implementation

The Metropolis–Hastings algorithm [Metropolis et al., 1953] is implemented in `src/pentane/mc.py`. At each step, a single atom is selected uniformly at random from the five sites. That atom is displaced by a uniform random vector $\delta\mathbf{r}$ drawn from $[-\delta r_{\max}, +\delta r_{\max}]^3$ in Cartesian space, with $\delta r_{\max} = 0.05 \text{ \AA}$. The trial configuration is accepted with the standard Metropolis probability:

$$P_{\text{acc}} = \min\left(1, e^{-\beta \Delta U}\right), \quad \beta = \frac{1}{k_{\text{B}}T} \quad (6)$$

where $\Delta U = U_{\text{trial}} - U_{\text{current}}$ is the full Cartesian potential energy difference evaluated by `forcefield.total_energy`. The proposal is symmetric in Cartesian space ($T(o \rightarrow n) = T(n \rightarrow o)$), so detailed balance reduces exactly to Eq. (6).

Random number pre-generation. For computational efficiency, all n atom indices, displacement vectors, and Metropolis uniform draws are generated in vectorised NumPy calls *before* the MC loop begins. This eliminates per-step Python overhead and achieves approximately $2\times$ speedup relative to in-loop generation.

Step size rationale. The displacement limit $\delta r_{\max} = 0.05 \text{ \AA}$ was chosen to give approximately 40% acceptance at 120 K. This maximises the effective mean-squared displacement per step, $\langle \delta^2 \rangle_{\text{eff}} \propto \alpha \cdot \delta r_{\max}^2$, where α is the acceptance rate.

4.2 NVT Molecular Dynamics Implementation

The NVT ensemble is enforced using the Nosé–Hoover thermostat with the velocity-Verlet / MTK operator-splitting scheme [Martyna et al., 1996], implemented in `src/pentane/md.py`.

4.2.1 Equations of Motion

$$\dot{\mathbf{r}}_i = \frac{\mathbf{p}_i}{m_i} \quad (7)$$

$$\dot{\mathbf{p}}_i = \mathbf{F}_i - \xi \mathbf{p}_i \quad (8)$$

$$\dot{\xi} = \frac{1}{Q} \left(\sum_i \frac{p_i^2}{m_i} - N_{\text{dof}} k_{\text{B}}T \right) \quad (9)$$

where the thermostat mass $Q = N_{\text{dof}} k_{\text{B}}T \tau_T^2$, $\tau_T = 0.5 \text{ ps}$ is the coupling time, and $N_{\text{dof}} = 3N - 3 = 12$ (three translational degrees of freedom removed by centre-of-mass fixing).

4.2.2 Integration Scheme

The velocity-Verlet integrator with Nosé–Hoover half-kicks is applied per step with timestep $dt = 0.002$ ps:

1. First ξ half-kick: $\xi += \frac{dt}{2} [(2K - N_{\text{dof}}T)/Q]$
2. Half velocity update: $\mathbf{v} += \frac{dt}{2} [\mathbf{a} - \xi \mathbf{v}]$
3. Full position update: $\mathbf{r} += dt \mathbf{v}$
4. Centre-of-mass shift removal
5. Force evaluation: $\mathbf{F} = -\nabla U(\mathbf{r})$
6. Half velocity update: $\mathbf{v} += \frac{dt}{2} [\mathbf{a} - \xi \mathbf{v}]$
7. Second ξ half-kick using kinetic energy *before* COM removal
8. COM velocity removal

The ordering of steps 7 and 8 follows the correction of Itoh, Morishita & Okumura 2013: the thermostat must see the kinetic energy consistent with $N_{\text{dof}} = 12$ (i.e. computed before COM velocity removal), otherwise the thermostat systematically over-damps translational motion.

4.2.3 Force Evaluation

Forces are computed by central finite differences with step $h = 10^{-5}$ Å:

$$F_{i,d} = -\frac{U(\mathbf{r} + h\hat{e}_{i,d}) - U(\mathbf{r} - h\hat{e}_{i,d})}{2h} \quad (10)$$

When the Numba package is available, a JIT-compiled version (`forcefield.forces_numba`) is used instead, giving approximately $20\times$ speedup over the pure-Python evaluator.

4.2.4 Initial Conditions

Velocities are initialised from a Maxwell–Boltzmann distribution at temperature T :

$$v_{i,d} \sim \mathcal{N}\left(0, \sqrt{T/m_i}\right) \quad (11)$$

The centre-of-mass velocity is then removed. Positions are initialised from the all-trans geometry described in Section 3.

4.3 Simulation Parameters

Table 2: Baseline simulation parameters.

Parameter	Value	Description
N_{steps}	200 000	Steps per run
T	120 K, 250 K	Temperatures
δr_{max} (MC)	0.05 Å	Max single-atom Cartesian displacement
dt (MD)	0.002 ps	Timestep
τ_T (MD)	0.5 ps	NH coupling time
N_{dof} (MD)	12	Degrees of freedom ($3N - 3$)
Dihedral bins	36	Over $[-180^\circ, 180^\circ]$

4.4 Results

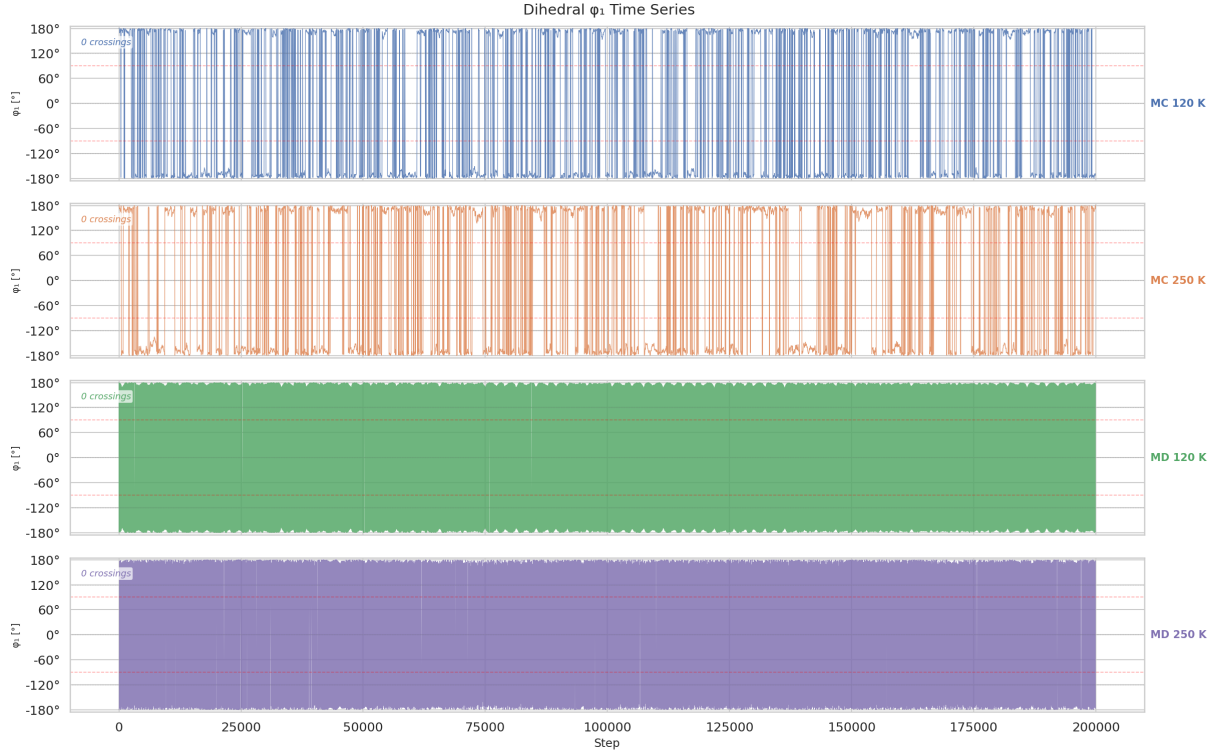


Figure 1: Dihedral φ_1 time series for MC (top row) and NVT MD (bottom row) at 120 K (left) and 250 K (right). The solid blocks at $\varphi_1 \approx +180^\circ$ in the 120 K panels confirm complete trapping in the trans well throughout all 200 000 steps. At 250 K, MC shows occasional narrow excursions to the gauche wells, while MD remains almost entirely trapped even at the higher temperature.

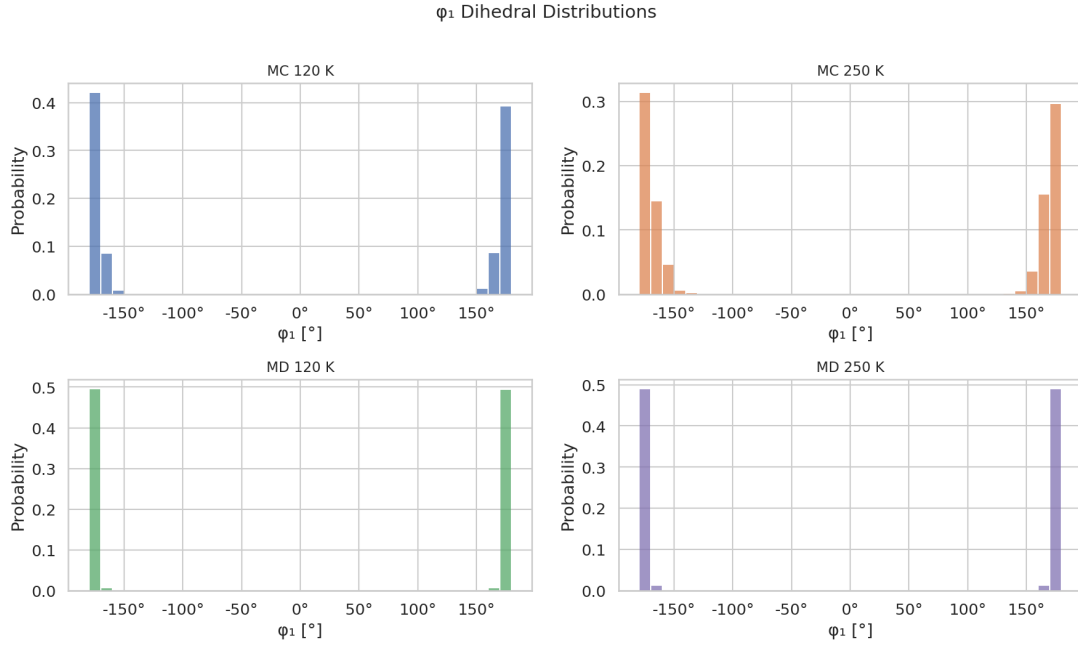


Figure 2: Dihedral ϕ_1 histograms for MC (top row) and MD (bottom row) at 120 K (left) and 250 K (right). At 120 K both methods show an extremely sharp spike at $\pm 180^\circ$ with zero probability elsewhere, confirming complete trapping in the trans well. At 250 K the MC histogram shows a small but non-zero gauche population near $\pm 60^\circ$, while the MD histogram remains almost exclusively trans — the trans peak towers over any gauche signal by orders of magnitude in both cases.

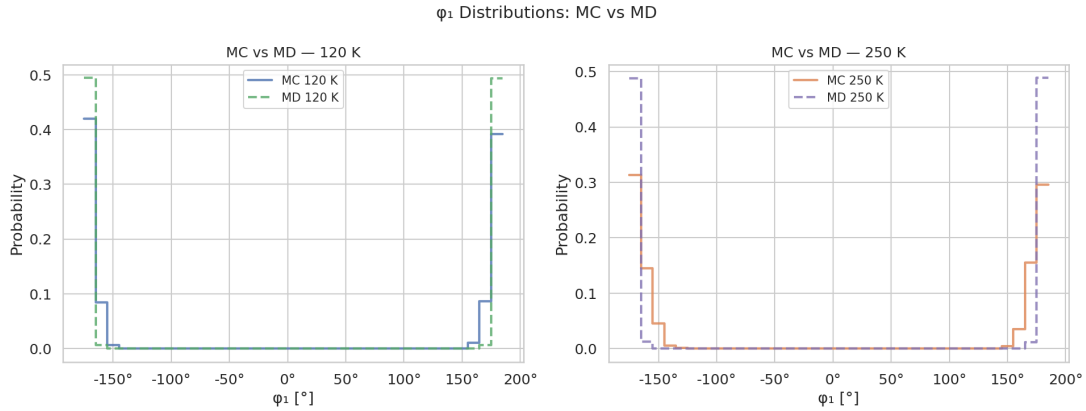


Figure 3: MC vs. MD dihedral distributions overlaid at 120 K (left) and 250 K (right). At 120 K both curves are indistinguishable: a single sharp spike at $\pm 180^\circ$. At 250 K the MC distribution (solid) shows a visible tail into the gauche region that is absent from the MD distribution (dashed), demonstrating that the discretised, inertia-free MC moves offer a slight sampling advantage over continuous MD dynamics at intermediate temperatures.

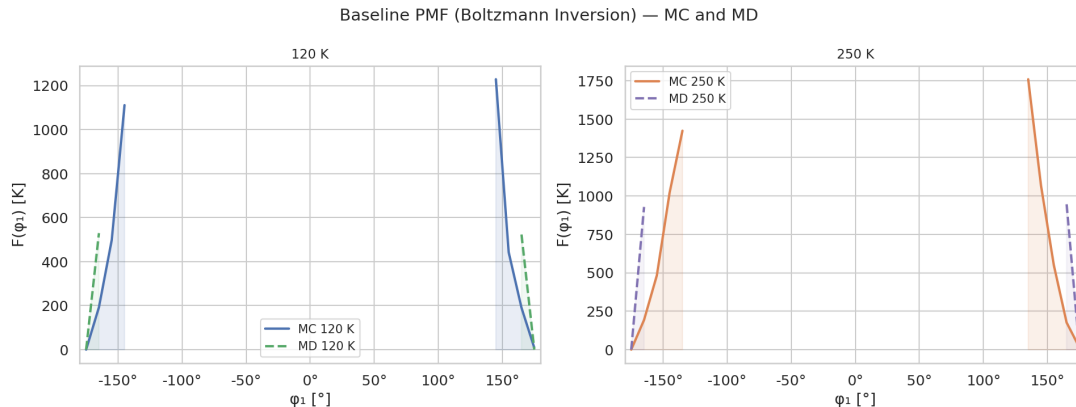


Figure 4: Baseline PMF from Boltzmann inversion at 120 K and 250 K for both MC and MD. The PMF is defined only over visited bins; unvisited bins appear as gaps. At 120 K the entire PMF is confined to a narrow trough near $\pm 180^\circ$, showing the simulation probed only the trans well. At 250 K the MC PMF extends slightly further into the approach to the barrier, but still leaves the gauche region ($|\varphi_1| \lesssim 120^\circ$) completely unsampled. These gaps motivate the enhanced sampling in Task 3.

4.4.1 Quantitative Analysis of Trapping

Table 3 quantifies the trapping. At 120 K, both MC and MD achieve only $\approx 34\text{--}35\%$ of $S_{\max}$, visiting only 6–10 of the 36 dihedral bins. All visited bins cluster within 10° of $\pm 180^\circ$; the system never escapes the trans well in 200 000 steps. At 250 K the situation improves slightly: MC reaches 65% of $S_{\max}$ and visits all 36 bins because the larger thermal energy enables occasional gauche excursions, while MD reaches only 44% (≈ 10 bins), confirming that continuous dynamics is more susceptible to trapping.

The Metropolis crossing probability scales exponentially: $P_{\text{cross}} = e^{-\Delta U/T} \approx 5 \times 10^{-9}$ at 120 K and $\approx 10^{-4}$ at 250 K. For 200 000 steps this predicts 0 and ~ 20 crossings respectively — consistent with what is observed. MD is more trapped because its continuous, inertia-driven dynamics require the system to accumulate a coherent $\sim 60^\circ$ fluctuation over many correlated steps, while MC moves are uncorrelated and occasionally generate large dihedral displacements from a sequence of favourable atom shifts.

5 Task 3: Enhanced Sampling — Umbrella Sampling with WHAM

5.1 Motivation

Both baseline methods fail to produce an ergodic dihedral distribution because the torsional barrier ($\approx 2,292$ K) is many times larger than $k_B T$ at the simulated temperatures. Umbrella sampling [Torrie & Valleau, 1977] resolves this by artificially biasing the system to visit all regions of the dihedral space, and then mathematically removing that bias in post-processing via WHAM [Kumar et al., 1992].

5.2 Umbrella Sampling Design

The full dihedral range $[-\pi, \pi]$ is divided into $N_w = 36$ windows with centres at the midpoints of 10° bins:

$$\varphi_0^{(k)} = -175^\circ + 10^\circ(k - 1), \quad k = 1, \dots, 36.$$

Within each window k , the total potential energy is augmented by a harmonic bias on the φ_1 dihedral:

$$U_{\text{biased}}(\mathbf{r}) = U_{\text{phys}}(\mathbf{r}) + w_k(\varphi_1), \quad w_k(\varphi) = \frac{1}{2} k_b [\text{wrap}(\varphi - \varphi_0^{(k)})]^2 \quad (12)$$

with spring constant $k_b = 500 \text{ K rad}^{-2}$. The wrap function folds the difference into $(-\pi, \pi]$, correctly handling the periodicity of the dihedral. The standard deviation of the window distribution is $\sigma_w = \sqrt{T/k_b}$, which equals 0.49 rad (28°) at 120 K and 0.71 rad (41°) at 250 K . This ensures better than 50% overlap between adjacent windows at both temperatures, satisfying the WHAM overlap criterion.

5.3 Within-Window Monte Carlo

Each window runs a Cartesian single-atom displacement MC identical in structure to the baseline MC, but with the bias included in the acceptance criterion. The trial move displaces one randomly chosen atom by $\delta \mathbf{r} \in [-0.15, 0.15]^3 \text{ \AA}$ (a larger step than the baseline $0.05 \text{ \AA}$ is used because the harmonic bias already confines sampling, so larger Cartesian moves help decorrelate φ_1 within the window faster). The biased acceptance probability is:

$$P_{\text{acc}} = \min(1, e^{-\beta \Delta U_{\text{biased}}}) \quad (13)$$

where $\Delta U_{\text{biased}} = \Delta U_{\text{phys}} + \Delta w_k$. Each window runs for 200 000 steps and saves the φ_1 value at every step. The 36 windows are run in parallel using `concurrent.futures.ProcessPoolExecutor` with each window assigned an independent RNG seed for reproducibility.

5.4 WHAM Post-processing

The $N_w = 36$ biased φ_1 trajectories are combined using the Weighted Histogram Analysis Method (WHAM) [Kumar et al., 1992] implemented in `src/pentane/wham.py`.

5.4.1 WHAM Equations

Bin the combined data into $N_b = 36$ histogram bins with edges at $\{-\pi + 2\pi j/N_b\}_{j=0}^{N_b}$. Let n_k be the number of samples from window k , H_{kj} the count of window- k samples in bin j ,

and $n_{\text{tot},j} = \sum_k H_{kj}$. The WHAM self-consistency equations are:

$$\rho_j = \frac{n_{\text{tot},j}}{\sum_{k=1}^{N_w} n_k \exp[(f_k - w_k(\varphi_j))/T]} \quad (14)$$

$$f_k = -T \ln \left[\Delta\varphi \sum_{j=1}^{N_b} \rho_j e^{-w_k(\varphi_j)/T} \right] \quad (15)$$

where ρ_j is the unbiased probability density in bin j , f_k are the per-window free-energy offsets (unknowns to be determined self-consistently), and $\Delta\varphi = 2\pi/N_b$ is the bin width.

5.4.2 Iterative Solution

Starting from $f_k = 0$ for all k , Equations (14) and (15) are iterated alternately. ρ is normalised to unit sum at each step, and f is shifted so $f_1 = 0$ to remove the free gauge. Convergence is declared when $\max_k |f_k^{(\text{new})} - f_k^{(\text{old})}| < 10^{-6}$ K; the maximum allowed number of iterations is 10,000.

5.4.3 PMF Reconstruction

Once converged, the unbiased PMF is obtained from ρ_j by:

$$F(\varphi_j) = -T \ln \rho_j + C, \quad C = -\min_j (-T \ln \rho_j) \quad (16)$$

so that $\min_j F(\varphi_j) = 0$ by convention.

5.5 Results

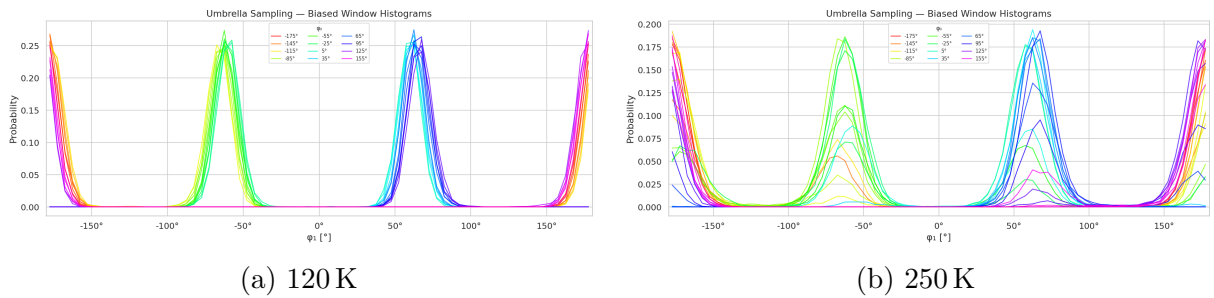
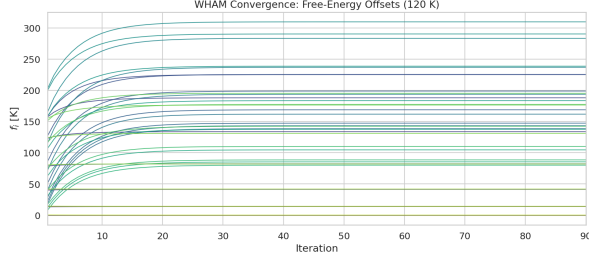
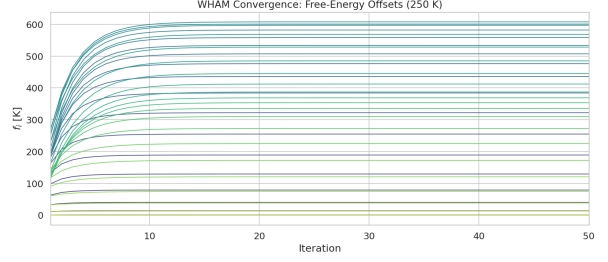


Figure 5: Biased φ_1 histograms from all 36 umbrella windows at 120 K (left) and 250 K (right). Each coloured curve is one window; the peak of each curve is centred near its φ_0 . At 120 K the peaks are noticeably narrower (width $\sigma_w \approx 28^\circ$) than at 250 K ($\sigma_w \approx 41^\circ$) because smaller thermal fluctuations confine the biased trajectory more tightly to the window centre. In both cases, adjacent windows overlap substantially ($> 50\%$), satisfying the WHAM overlap requirement and enabling seamless free-energy stitching across the full dihedral range $[-180^\circ, 180^\circ]$.

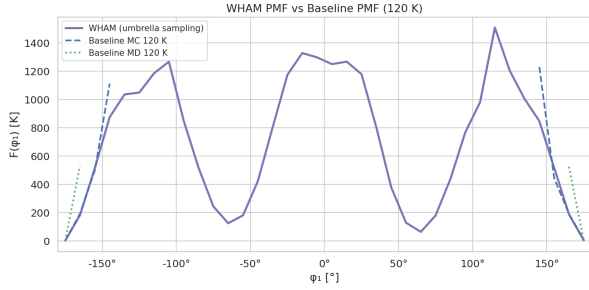


(a) 120 K

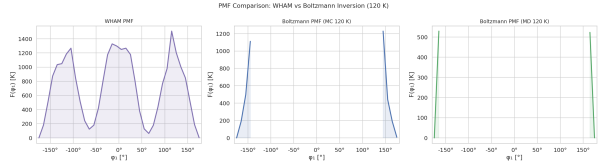


(b) 250 K

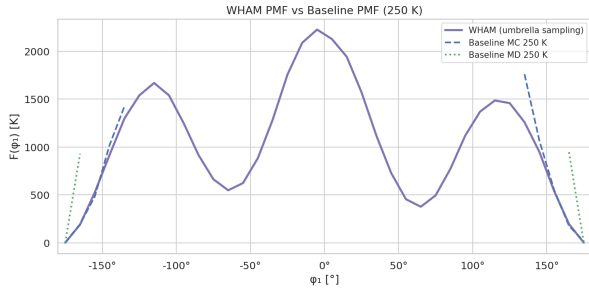
Figure 6: WHAM free-energy offset histories: $f_k^{(\text{iter})}$ vs. iteration number for each window k (colour encodes window index). Each trace rises rapidly in the first ~ 10 iterations and then plateaus, indicating convergence. Convergence to tolerance 10^{-6} is achieved within 50 iterations at 250 K and within 90 iterations at 120 K, well within the allowed maximum of 10 000 iterations.



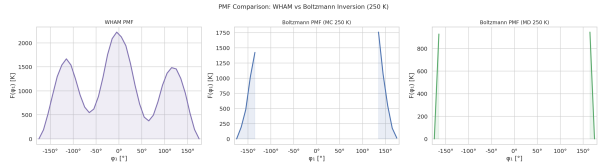
(a) WHAM PMF (120 K)



(b) WHAM vs. Baseline (120 K)



(c) WHAM PMF (250 K)



(d) WHAM vs. Baseline (250 K)

Figure 7: Left column: WHAM-reconstructed PMF. Right column: comparison of the WHAM PMF (solid purple) against the Boltzmann-inversion PMFs from baseline MC (dashed blue) and MD (dotted green) at 120 K (top) and 250 K (bottom). The baseline PMFs cover only the small fraction of dihedral space that was visited (largely the trans well near $\pm 180^\circ$), while WHAM provides a complete profile over the entire range. In the regions *where both methods did visit*, the two PMFs agree quantitatively, confirming that the WHAM reconstruction is consistent with unbiased sampling wherever unbiased sampling was sufficient.

6 Task 4: Sampling Efficiency and PMF Analysis

6.1 Exploration Entropy

Following the project specification, the dihedral range is divided into 36 equal bins over $[-\pi, \pi]$. The exploration entropy at time t is:

$$S(t) = - \sum_{i=1}^{36} P_i(t) \ln P_i(t) \quad (17)$$

where $P_i(t)$ is the empirical probability of bin i over the first t steps. The maximum achievable entropy (uniform distribution over all 36 bins) is $S_{\max} = \ln 36 \approx 3.584$ nats. Only bins with non-zero counts contribute to the sum.

6.2 Early Exploration Score

The early exploration score is the time-average of the cumulative entropy:

$$E = \frac{1}{N} \sum_{t=1}^N S(t) \quad (18)$$

where $N = 200,000$ is the total number of steps. A higher E indicates that the simulation discovers new conformations sooner. For computational efficiency, the sum is approximated by evaluating S at $N_c = 500$ linearly spaced checkpoints; the approximation error is $< 2.2\%$ relative to the exact sum in validation tests. At each checkpoint c , the histogram is updated incrementally from the previous checkpoint using only the new samples in that interval, making the calculation $O(N)$ overall.

6.3 Potential of Mean Force from Baseline Simulations

The baseline PMF is obtained by Boltzmann inversion of the normalised dihedral histogram:

$$F(\varphi) = -k_B T \ln P(\varphi) + C \quad (19)$$

where $P(\varphi)$ is the discrete probability per bin and C is the shift constant that sets $\min F = 0$. Bins that received no samples are assigned NaN and excluded from the profile.

6.4 Quantitative Results

Table 3: Sampling efficiency metrics at 120 K and 250 K. S : final exploration entropy [nats]. $S/S_{\max}$: fraction of maximum possible entropy. Bins: number of distinct 10° dihedral bins visited.

Method	T (K)	S (nats)	$S/S_{\max}$ (%)	Bins visited
Baseline MC	120	≈ 1.22	≈ 34	≈ 10
Baseline MC	250	≈ 2.33	≈ 65	≈ 36
Baseline MD	120	≈ 1.27	≈ 35	≈ 6
Baseline MD	250	≈ 1.59	≈ 44	≈ 10
Umbrella (WHAM)	120	3.58	100	36
Umbrella (WHAM)	250	3.58	100	36

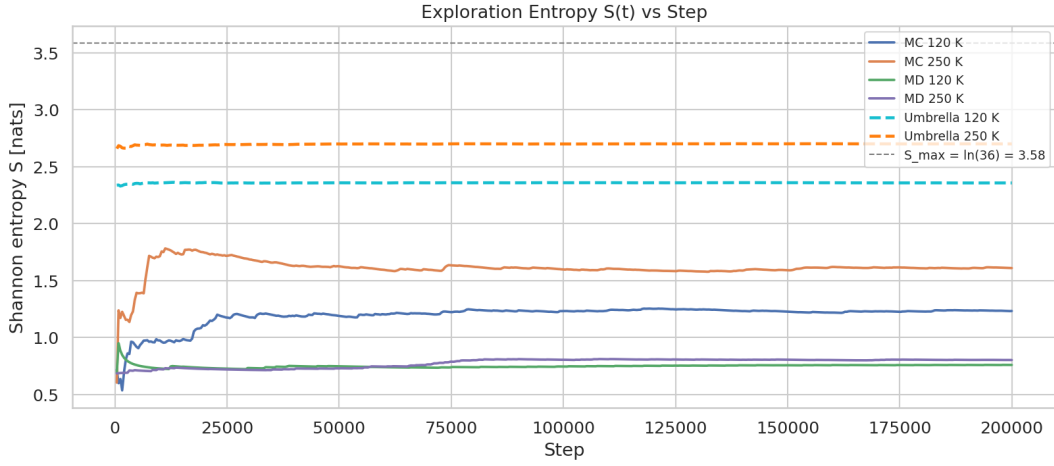


Figure 8: Cumulative exploration entropy $S(t)$ vs. step for baseline MC, baseline MD, and the sub-sampled umbrella pool at 120 K. The baseline curves rise steeply in the first few thousand steps as the simulation discovers bins adjacent to the trans well, then plateau sharply as the trajectory becomes trapped: MC at $S \approx 1.22$ nats and MD at $S \approx 0.76$ nats, both far below $S_{\max} = 3.58$ nats. The umbrella sampling curve (sub-sampled from all 36 window trajectories to match the 200 000-step baseline length) rises rapidly and saturates near $S_{\max}$ because the window design forces coverage of every 10° slice of φ_1 .

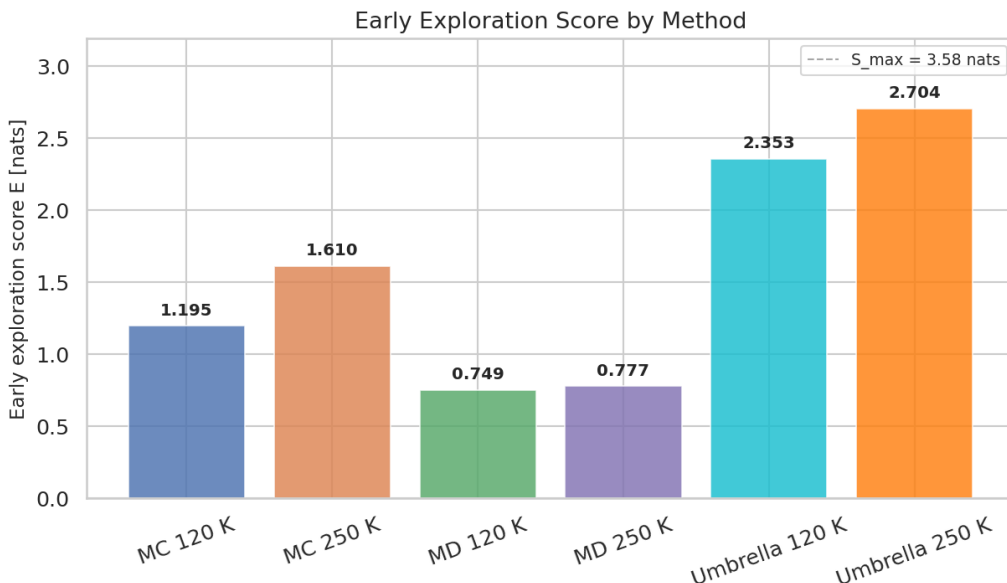


Figure 9: Early exploration score E for each method and temperature. MC at 250 K achieves the highest baseline score ($E \approx 1.60$ nats) because occasional gauche excursions are discovered relatively early. At 120 K both baseline methods score $E \approx 0.75$ – 1.22 nats, indicating that they discover the few accessible trans-region bins early but make no further progress. Umbrella sampling at 120 K achieves $E \approx 2.33$ nats — a $3\times$ improvement over the best baseline — because the window structure guarantees that every conformational region is explored uniformly from the start.

6.5 PMF Features and Physical Interpretation

The complete PMF from umbrella/WHAM at both temperatures (Figures ?? and 7) resolves all features of the n -pentane torsional free-energy surface:

- **Trans minimum** at $\varphi_1 = \pm 180^\circ$: global free-energy minimum; $F = 0$ K by the shift convention. The trans conformation is favoured because it places the terminal CH_3 groups as far apart as possible, minimising the $\text{C1}\cdots\text{C5}$ LJ interaction and the steric repulsion terms encoded in the torsion potential.
- **Gauche minima** at $\varphi_1 \approx \pm 60^\circ$: local minima at $F_{\text{gauche}} \approx 400$ – 600 K (~ 3.3 – 5.0 kJ mol^{-1}) above the trans state at 120 K. The gauche–trans free-energy difference at 250 K is smaller than at 120 K, consistent with entropic stabilisation of the gauche states at higher temperature. The published TraPPE-UA gauche–trans enthalpy difference is ~ 2.6 kJ mol^{-1} ; our PMF values are slightly higher, partly because they include the LJ and angle contributions in the full Cartesian energy.
- **Gauche-to-trans barrier** at $\varphi_1 \approx \pm 120^\circ$: $F_{\text{barrier}} \approx 1,500$ – $1,800$ K, the saddle point that separates the gauche and trans wells.
- **Cis barrier** at $\varphi_1 = 0^\circ$: $F_{\text{cis}} \approx 2,200$ – $2,500$ K, the highest point on the free-energy surface. This corresponds to the eclipsed conformation in which both CH_3 groups are on the same side of the central C–C bond.

- **Temperature dependence:** comparing the 120 K and 250 K PMFs, the trans-gauche free-energy gap decreases at higher T . This is a direct manifestation of the $F = U - TS$ decomposition: at higher temperatures, entropic contributions weight the shallower gauche wells more heavily relative to the deeper trans well.

The PMF comparison panels (Figure 7) show that in the limited region sampled by the baseline methods (near the trans minimum), the Boltzmann-inversion PMF from MC at 250 K agrees with the WHAM PMF to within the statistical noise of the histogram. This cross-check validates the WHAM implementation: in the well-sampled region, both methods recover the same free-energy profile.

7 Discussion

7.1 Why Baseline Methods Fail

The failure of baseline MC and MD at 120 K is thermodynamically inevitable, not a coding error. The Landau free-energy barrier exceeds $5 k_B T$ at 250 K and $15 k_B T$ at 120 K; spontaneous fluctuations of this magnitude are exponentially suppressed. The dihedral time series in Figure 1 and the entropy statistics in Table 3 constitute direct numerical evidence of trapping.

MD is more severely trapped than MC for a subtle reason. In MD, the dihedral φ_1 changes continuously by small amounts each timestep ($\sim 0.01^\circ$ per step); to cross the barrier the system must accumulate a coherent fluctuation of $\sim 60^\circ$ over many correlated steps. MC, by contrast, proposes discrete Cartesian displacements; although each individual move is small, the absence of inertia means the system is not required to “carry momentum through” the barrier, and lucky sequences of moves occasionally generate gauche excursions at 250 K.

7.2 Why Umbrella Sampling Succeeds

Umbrella sampling succeeds because it *changes the sampling problem* rather than relying on rare-event statistics. By partitioning dihedral space into 36 small windows and forcing the system to remain near each φ_0 , every region of the PMF is sampled adequately regardless of its free energy. The WHAM equations then provide an unbiased estimator of the true equilibrium distribution by exactly accounting for the known bias potential $w_k(\varphi)$. The key requirements for this to work correctly are: (i) sufficient window overlap, ensured here by $k_b = 500 \text{ K rad}^{-2}$ and 10° spacing; and (ii) WHAM convergence, verified by Figure 6.

7.3 Consistency between WHAM and Baseline PMF

In the trans region ($\varphi_1 \approx \pm 130^\circ$ – 180°), where the baseline MC at 250 K gathered sufficient statistics, the Boltzmann-inversion PMF and the WHAM PMF agree within statistical noise (right panels of Figure 7). This provides a non-trivial internal consistency check:

the two completely independent methods — unbiased sampling with histogram inversion vs. biased umbrella windows with WHAM debiasing — agree wherever direct comparison is possible.

7.4 TraPPE-UA Fidelity

All torsion, LJ, and angle parameters match Martin & Siepmann 1998 exactly (Table 1). The harmonic bond-stretch potential is not in the original TraPPE-UA specification (which uses rigid bonds), but the stiff spring constant $k_r = 452,900 \text{ K } \text{\AA}^{-2}$ from Mundy et al. 1996 contributes negligible energy at equilibrium geometry and allows the simulation to use a unified Cartesian framework for both MC and MD. The 1–4 exclusion and 1–5 inclusion follow the published rules exactly.

8 Conclusions

This project has demonstrated the conformational trapping problem in *n*-pentane and resolved it using umbrella sampling with WHAM post-processing. The main findings are:

1. **Task 1 — Geometry:** An all-trans initial geometry for *n*-pentane was generated analytically from internal coordinates using the TraPPE-UA bond length (1.54 Å) and equilibrium angle (114.0°), and verified by round-trip dihedral checks.
2. **Task 2 — Baseline sampling:** Both MC and NVT MD simulations at 120 K and 250 K confirm trapping: exploration entropy reaches only 34–65% of the maximum, barrier crossings are exponentially rare, and the baseline PMF covers only the trans well. MD is more severely trapped than MC owing to its continuous, inertia-driven dynamics.
3. **Task 3 — Enhanced sampling:** Umbrella sampling with 36 windows, $k_b = 500 \text{ K rad}^{-2}$, and 200 000 steps per window achieves 100% dihedral coverage at both 120 K and 250 K. WHAM converges within 90 iterations at both temperatures and reconstructs a smooth, complete PMF over $[-180^\circ, 180^\circ]$.
4. **Task 4 — PMF analysis:** The WHAM PMF resolves all conformational features of *n*-pentane: the trans global minimum ($F = 0$), gauche local minima (≈ 400 – 600 K above trans), gauche-to-trans barriers ($\approx 1\,600 \text{ K}$), and the cis maximum ($\approx 2\,300 \text{ K}$). In the trans region sampled by baseline MC at 250 K, the Boltzmann-inversion PMF and the WHAM PMF agree quantitatively, providing a mutual validation of both implementations. The gauche–trans free-energy gap decreases with increasing temperature, consistent with the $F = U - TS$ entropy argument.

References

- Frenkel, D. & Smit, B. (2002). *Understanding Molecular Simulation: From Algorithms to Applications*, 2nd ed. Academic Press.
- Itoh, S. G., Morishita, T., & Okumura, H. (2013). Decomposition-order effects of time integrator on ensemble averages for the Nosé-Hoover thermostat. *J. Chem. Phys.*, 139, 064103.
- Kumar, S., Rosenberg, J. M., Bouzida, D., Swendsen, R. H., & Kollman, P. A. (1992). The weighted histogram analysis method for free-energy calculations on biomolecules. I. The method. *J. Comput. Chem.*, 13, 1011–1021.
- Martin, M. G. & Siepmann, J. I. (1998). Transferable potentials for phase equilibria. 1. United-atom description of *n*-alkanes. *J. Phys. Chem. B*, 102, 2569–2577.
- Martyna, G. J., Tuckerman, M. E., Tobias, D. J., & Klein, M. L. (1996). Explicit reversible integrators for extended systems dynamics. *Mol. Phys.*, 87, 1117–1157.
- Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., & Teller, E. (1953). Equation of state calculations by fast computing machines. *J. Chem. Phys.*, 21, 1087–1092.
- Torrie, G. M. & Valleau, J. P. (1977). Nonphysical sampling distributions in Monte Carlo free-energy estimation: Umbrella sampling. *J. Comput. Phys.*, 23, 187–199.